Healthcare Infectious Disease Simulation

A Monte Carlo Analysis of Protection, Population Survival, and Economic Preservation

Excel-based stochastic dynamic simulation | 100 periods | 1,000 simulated villages

Portfolio Project Report

Keman Xiang

kemanxiang005@gmail.com

Feb 2024

Source: MGT239 Simulation for Business Course Project

Abstract

This project develops an Excel-based stochastic dynamic simulation to evaluate the **spread of an infectious disease** in a closed village and to **compare two policy scenarios**: **no protection** and universal use of an **infection-protection** measure. The village begins with 100 residents, including 95 susceptible and 5 infected individuals. In each period, an infected resident faces a 10% mortality probability, encounters a particular susceptible with probability 5%, and—without protection—transmits infection conditional on an encounter with probability 50%. **Protection** reduces the conditional infection probability to 10% while leaving the encounter probability unchanged. The model follows the **susceptible**, **infected**, and **cumulative-death** states for **100 periods** and replicates the process across **1,000 simulated villages**.

The Monte Carlo results show a large difference in population survival. The estimated probability of **complete population die-out** is **99.3% without protection** (95% CI: 98.783%–99.817%) and **56.0% with protection** (95% CI: 52.922%–59.078%). **Disease extinction** by period 100 is high in both scenarios—99.3% without protection and 99.6% with protection—while **cumulative infection** remains nearly universal: 100.0% versus 99.352%. The principal economic difference arises from preserving the living population over the simulation horizon. **Average cumulative tax revenue** increases from $12.610 million to $17.141 million per simulated village, a gain of approximately $4.531 million, or 35.9%, under protection. Convergence diagnostics show that the main estimates stabilize as the replication count approaches 1,000.

The analysis demonstrates a broader decision-analytics lesson: evaluating a stochastic intervention only by its terminal disease status can obscure economically and operationally important path effects. Even when disease extinction is nearly certain in both scenarios, **a reduction in transmission can materially alter the trajectory of mortality and preserve a larger economic base**. The model is intentionally stylized and is best interpreted as a simulation study of **dynamic risk and policy comparison** rather than as an epidemiological forecast.

Table of Contents

Abstract 2
Table of Contents 3
1. Problem Background and Decision Context 4
2. Theoretical and Modeling Framework 4
2.1 State Representation 4
2.2 Transmission Probability 4
2.3 Infected Survival and State Transition 5
2.4 Monte Carlo Output Analysis 5
3. Excel Simulation Design and Project Workflow 5
3.1 Input Layer 5
3.2 Sample-Path Engine 6
3.3 Replication and Statistical Output 6
4. Results 7
4.1 Main Monte Carlo Estimates 7
4.2 Population Survival 8
4.3 Disease Extinction and Cumulative Infection 8
4.4 Economic Outcome: Cumulative Tax Revenue 8
5. Simulation Convergence and Reliability 9
6. Interpretation 11
6.1 Path Dependence Matters More Than Terminal Status 11
6.2 Protection Changes the Dynamic Feedback Loop 11
6.3 Why the Infection-Rate Result Is Not a Contradiction 12
7. Managerial and Business Implications 12
8. Limitations and Extensions 12
9. Conclusion 13
References 14
Appendix A. Key Workbook Outputs Used in This Report 14

1. Problem Background and Decision Context

Infectious-disease control is a dynamic decision problem because current infections affect future exposure, future population composition, and downstream economic outcomes. A policy that reduces transmission does not merely change the probability of a single infection; it changes the state of the system that generates subsequent infections. This feedback makes simulation particularly useful when closed-form intuition is insufficient.

The setting considered here is a **remote village** with a fixed initial population of **100**. At the beginning of period 1, 95 residents are susceptible and 5 are infected. During each period, each infected resident has a **10% probability of dying** by the end of that period. An infected resident **encounters** any particular **susceptible** with probability **5%**; conditional on such an encounter, **infection** occurs with probability **50%**. The model tracks the village for **100 periods** and asks four operational questions: **whether the entire population dies out, whether the disease dies out, how much of the original population is ever infected,** and **how the population trajectory affects cumulative tax revenue**.

The **policy scenario** introduces a **protection** measure used during encounters. The intervention does not change the encounter probability. Instead, it reduces the conditional probability of infection from 50% to **10%**. This distinction is important because it isolates the intervention as a transmission-risk reduction rather than a behavioral reduction in contact. Each living resident contributes **$10,000** in **tax revenue** per period, allowing the health-state simulation to be translated into a stylized economic consequence.

The decision problem is therefore not simply whether protection causes the disease to disappear. It is **whether lowering transmission changes the distribution of population survival and cumulative economic output across many possible stochastic epidemic paths**.

2. Theoretical and Modeling Framework

2.1 State Representation

The system is represented by three state variables. S_t denotes the number of susceptible residents in period t, I_t denotes the number of infected residents, and D_t denotes cumulative deaths. The initial state is $S_1 = 95$, $I_1 = 5$, and $D_1 = 0$. Because the population is closed and the model contains no births, migration, or recovery state, the accounting identity $S_t + I_t + D_t = 100$ holds throughout the simulation.

This state-based representation is related to compartmental epidemic modeling in that individuals move between disease states, but the implementation here is deliberately narrower than a conventional SIR model: the 'removed' population is specifically cumulative deaths rather than a combined recovered/removed class. This distinction is maintained throughout the workbook and report.

2.2 Transmission Probability

Without protection, the probability that one infected resident infects one particular susceptible during a period is $0.05 \times 0.50 = 0.025$. Therefore, the probability that the infected resident does not infect that susceptible is 0.975. If I_t residents are infected, the probability that a particular susceptible avoids infection from all current infectives is:

$$q_t = (1 - 0.05 \times 0.50)^{It} = 0.975^{It}$$

Given S_t susceptible residents, the number remaining susceptible in the next period is modeled as a binomial random variable:

$$S_{t+1} \sim \text{Binomial}(S_t, q_t)$$

Under **protection**, the conditional infection probability falls from 0.50 to 0.10. The per-pair transmission probability becomes 0.05 × 0.10 = 0.005, so the corresponding probability of avoiding infection is:

$$q_t^{P} = (1 - 0.05 \times 0.10)^{It} = 0.995^{It}$$

2.3 Infected Survival and State Transition

Each currently infected resident survives the period with probability 0.90. The next-period infected population therefore combines two stochastic components: **surviving current infectives** and **newly infected residents**. In the workbook, the surviving infected count is generated from a binomial distribution and newly infected residents are computed as $S_t - S_{t+1}$. Cumulative deaths are then obtained from the population identity.

The nonlinear feedback is central to the model. A reduction in infections in one period reduces I_t in subsequent periods, which raises the probability that each remaining susceptible avoids infection. Consequently, the intervention can change the entire trajectory rather than only producing a one-period proportional effect.

2.4 Monte Carlo Output Analysis

One simulated village produces only one possible realization of the stochastic process. The workbook therefore repeats the **100-period** sample-path calculation across **1,000 villages** using Excel Data Tables. Each replication records four outcome statistics: an indicator for **disease die-out**, an indicator for **complete population die-out**, the **cumulative infection rate**, and **cumulative tax revenue**.

The output analysis follows standard Monte Carlo practice: probability outcomes are estimated by averaging indicator variables across replications, continuous outcomes are summarized by their sample means, and 95% confidence intervals are used to communicate Monte Carlo sampling uncertainty. Simulation-methodology literature emphasizes that a single run should not be treated as the answer and that independent replications and confidence intervals are central to credible output analysis [1,2]. Reporting both summary measures and graphical diagnostics also helps users distinguish the estimated effect from random simulation noise [1].

3. Excel Simulation Design and Project Workflow

3.1 Input Layer

The Excel model begins with a compact input block containing the initial state and scenario parameters: S_1 = 95, I_1 = 5, total population = 100, encounter probability = 0.05, no-protection conditional infection probability = 0.50, protection conditional infection probability = 0.10, mortality probability = 0.10, and

Monte Carlo sample size N = 1,000. Keeping these quantities explicit makes the model auditable and allows the scenario difference to be traced to a single transmission parameter.

3.2 Sample-Path Engine

The calculation sheet constructs two parallel 100-period sample paths: one for **no protection** and one for **protection**. For each period, the sheet calculates the probability of remaining uninfected, draws the next susceptible population using BINOM.INV and RAND(), draws infected survivors using a binomial random variable, and updates cumulative deaths. The two scenarios use the same state-transition logic; only the conditional infection parameter differs.

The current workbook also illustrates why individual paths should not be overinterpreted. In the displayed no-protection path, the susceptible population reaches zero by period 6 and cumulative deaths eventually reach 100. In the displayed protection path, transmission is slower, but that particular realization also eventually reaches 100 cumulative deaths. These are two valid stochastic paths, not estimates of policy performance. The policy conclusion comes from the distribution across 1,000 replications.

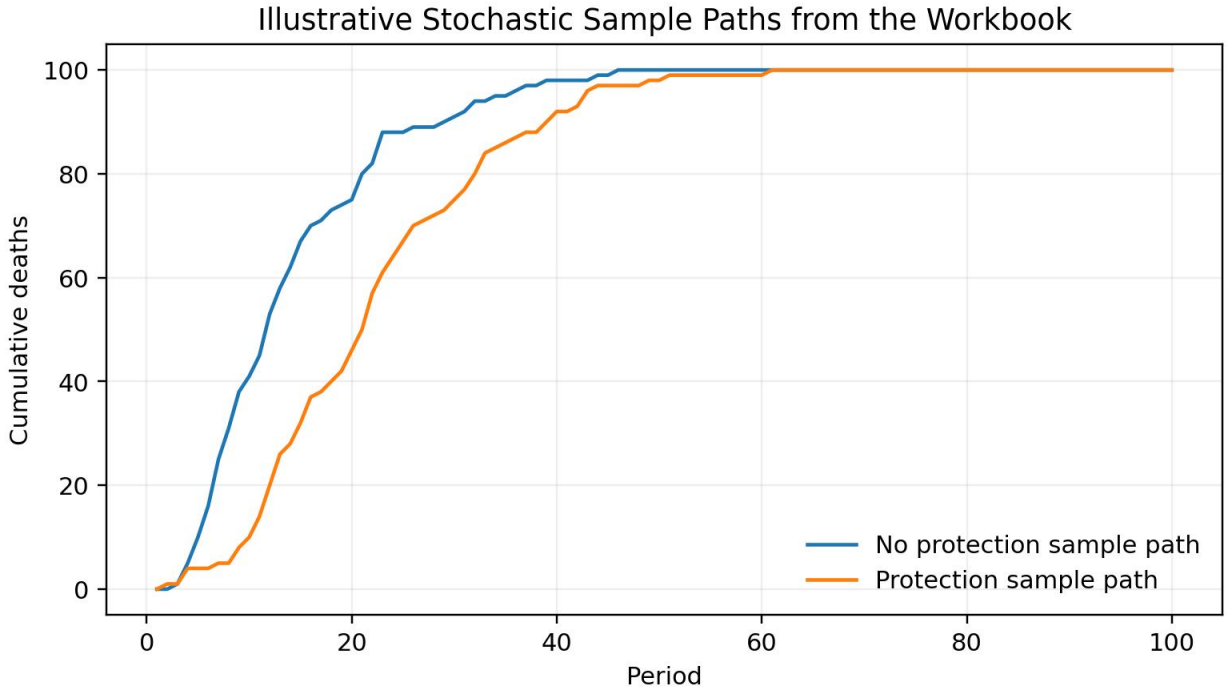


Figure 1. Illustrative cumulative-death sample paths stored in the final workbook.

3.3 Replication and Statistical Output

The workbook's STAT block converts the terminal state of each sample path into analysis-ready outputs. **Disease die-out** equals one when no infected residents remain at period 100. **Population die-out** equals one when cumulative deaths equal the original population of 100. The **cumulative infection rate** measures the share of the original population that has ever been infected by the horizon, while **cumulative tax** sums the tax contribution of living susceptible and infected residents over the 100 periods.

Excel Data Tables then replicate the stochastic model 1,000 times for each scenario. The OUTPUT blocks calculate scenario means or probabilities and 95% confidence intervals. A separate convergence sheet recomputes running estimates using the first 50, 100, 250, 500, 750, and 1,000 replications.

4. Results

4.1 Main Monte Carlo Estimates

Outcome	No Protection	Protection	Absolute Change	95% CI / Notes
Population die-out probability	99.3%	**56.0%**	−43.3 pp	No protection: 98.783%–99.817%; Protection: 52.922%–59.078%
Disease die-out probability	99.3%	99.6%	+0.3 pp	No protection: 98.783%–99.817%; Protection: 99.209%–99.991%
Average cumulative infection rate	100.0%	99.352%	−0.648 pp	No protection: 100.0%–100.0%; Protection: 99.293%–99.411%
Average cumulative tax	$12.610M	**$17.141M**	+$4.531M	No protection: $12.549M–$12.670M; Protection: $17.054M–$17.228M

Table 1. Final 1,000-village Monte Carlo estimates from the portfolio workbook.

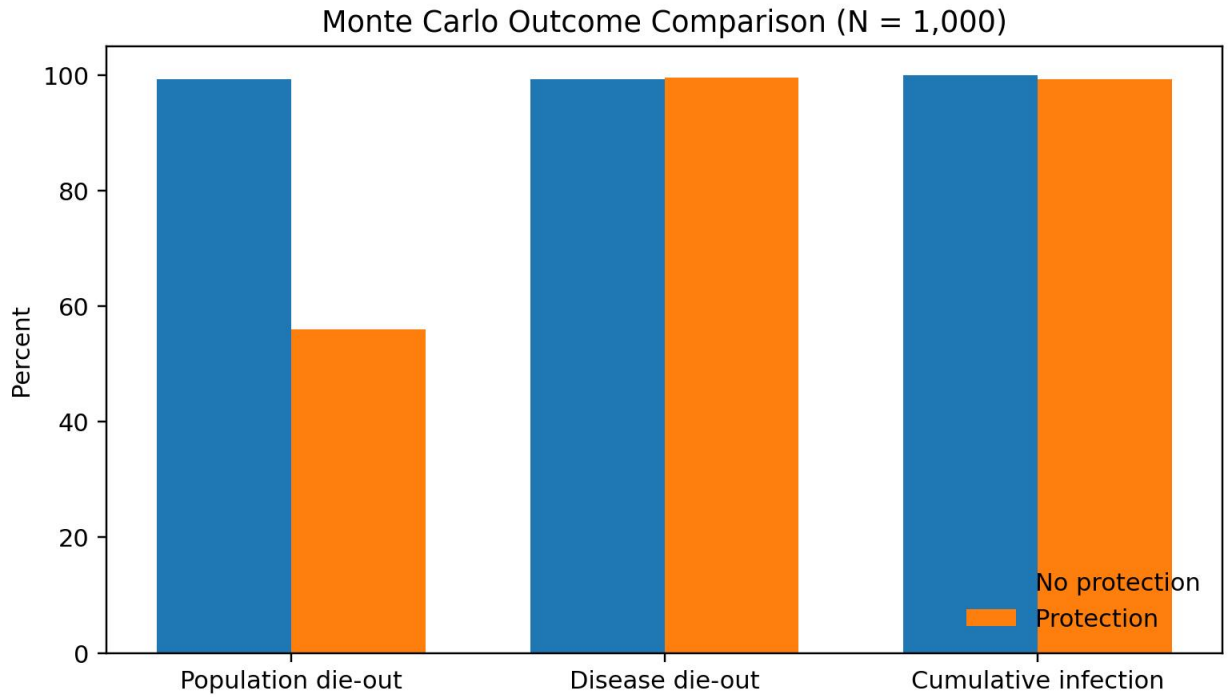


Figure 2. Comparison of population die-out, disease die-out, and cumulative infection outcomes across 1,000 simulated villages.

4.2 Population Survival

The largest policy effect appears in complete population die-out. **Without protection**, **99.3%** of simulated villages experience complete population loss by period 100. **Under protection**, the estimated probability falls to **56.0%**. This is a reduction of 43.3 percentage points, or approximately 43.6% relative to the no-protection estimate. The confidence intervals are widely separated: 98.783%–99.817% without protection versus 52.922%–59.078% with protection.

This result shows that **lowering the conditional transmission probability materially changes population survival even though the intervention does not eliminate risk**. In the stylized model, protection moves the system from near-certain complete population loss to an outcome in which complete loss still occurs in slightly more than half of simulated villages. The intervention is therefore **risk-reducing** rather than risk-eliminating.

4.3 Disease Extinction and Cumulative Infection

Disease extinction is estimated at 99.3% without protection and 99.6% with protection, a difference of only 0.3 percentage points. Read in isolation, this metric could suggest that the two scenarios are nearly equivalent. The population-survival results demonstrate why that interpretation would be incomplete. A disease can be extinct at the terminal horizon because transmission has ended after severe mortality; terminal disease status does not describe the path by which extinction occurred.

The cumulative infection results reinforce the severity of the parameterized setting. The average cumulative infection rate is 100.0% without protection and 99.352% with protection. Protection reduces the estimated cumulative infection burden by 0.648 percentage points, but infection remains nearly universal. Thus, within the model's assumptions, the principal benefit of protection is not that most residents avoid infection over the full 100-period horizon. Rather, **the lower transmission probability slows and alters the epidemic path sufficiently to preserve more residents for longer and to reduce the probability of complete population loss**.

4.4 Economic Outcome: Cumulative Tax Revenue

The workbook translates population preservation into a stylized fiscal metric. Each living resident contributes $10,000 per period. Because the Excel formulas express this as 10 in thousands of dollars, the average cumulative tax values of 12,609.66 and 17,141.03 correspond to approximately $12.610 million and **$17.141 million** per simulated village over the 100-period horizon.

Protection increases average cumulative tax revenue by 4,531.37 thousand dollars, or approximately $4.531 million. Relative to the no-protection scenario, this is a **35.9% increase**. The 95% confidence intervals are approximately $12.549–$12.670 million without protection and $17.054–$17.228 million with protection. The separation between these intervals is large relative to Monte Carlo uncertainty in this run.

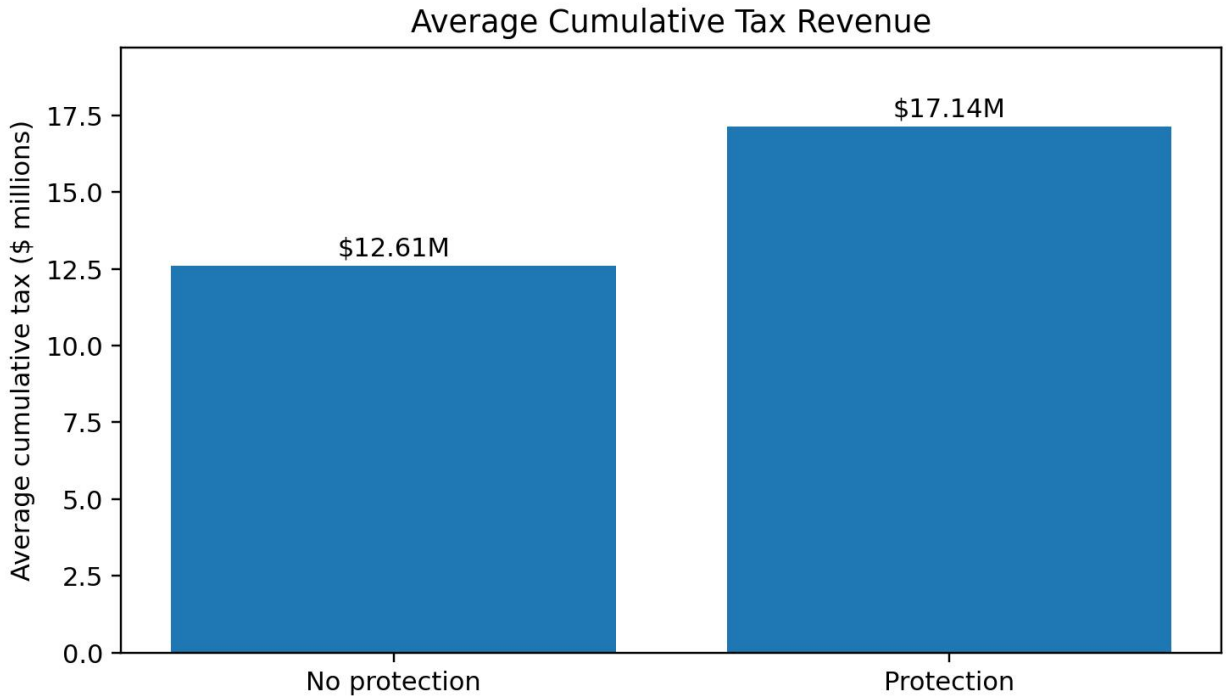


Figure 3. Average cumulative tax revenue across 1,000 simulated villages. Monetary values are converted from the workbook's $000s units to millions of dollars for readability.

5. Simulation Convergence and Reliability

A Monte Carlo estimate should become less sensitive to additional replications as the sample size increases. The workbook therefore evaluates running estimates at N = 50, 100, 250, 500, 750, and 1,000. This diagnostic does not prove that the model is realistic; rather, it checks whether the numerical estimates from the specified stochastic model are stabilizing.

For population die-out, the no-protection running estimate moves from 100.0% at N = 50 to 99.0%, 98.8%, 99.0%, 99.2%, and finally 99.3% at N = 1,000. The protection estimate moves from 58.0% to 56.0%, 58.0%, 55.8%, 55.6%, and 56.0%. The late-stage changes are small relative to the 43.3-percentage-point scenario difference.

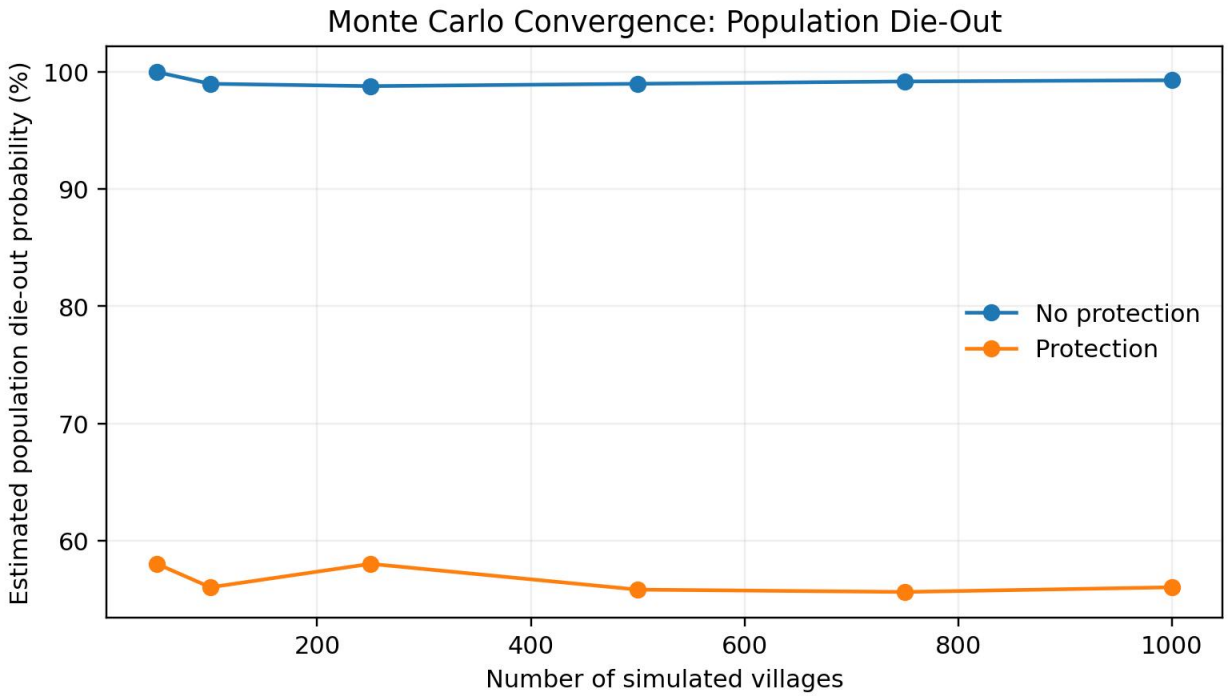


Figure 4. Running Monte Carlo estimates for complete population die-out. Both scenario estimates show increasing stability as the number of simulated villages approaches 1,000.

The tax outcome also stabilizes. The no-protection estimate is $12.727 million at N = 50 and $12.610 million at N = 1,000. The protection estimate moves from $16.862 million at N = 50 to $17.141 million at N = 1,000. Because tax is a continuous path-dependent outcome, it shows more visible finite-sample variation than the near-boundary probability measures, but the scenario separation remains large throughout the convergence sequence.

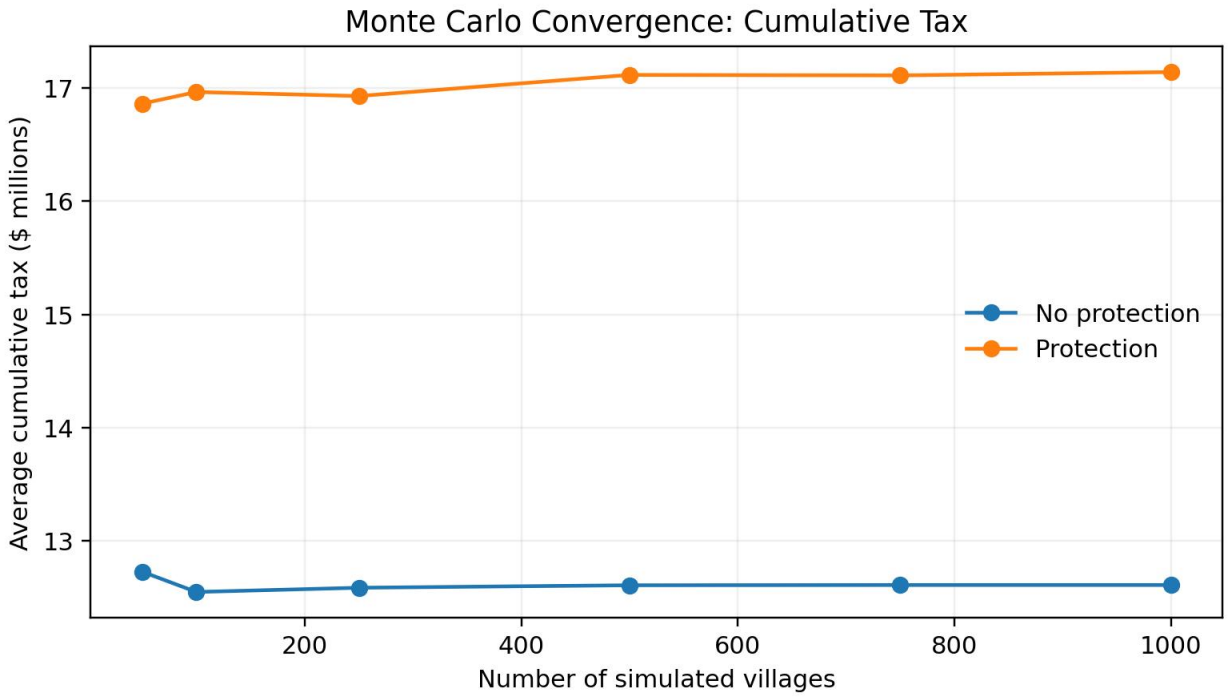


Figure 5. Running estimates of average cumulative tax revenue. The protection scenario remains economically higher throughout the displayed replication sequence.

The convergence evidence supports the use of N = 1,000 as a reasonable simulation sample for this portfolio analysis. More replications could further reduce Monte Carlo error, but the central scenario conclusions in the current workbook are not driven by the earliest, smallest replication counts.

6. Interpretation

6.1 Path Dependence Matters More Than Terminal Status

The most important analytical result is the divergence between disease extinction and population survival. **Disease extinction is almost certain in both scenarios, yet population die-out differs dramatically**. This occurs because terminal disease status discards information about the trajectory. Two villages can both have $I_{100} = 0$ while experiencing very different histories of infection and mortality.

For decision analysis, this means that **a terminal success metric can be misleading when the path itself carries costs**. In healthcare, operations, finance, and other stochastic systems, managers often care not only about where the system ends but also about cumulative losses, resource consumption, survival, and economic activity along the way.

6.2 Protection Changes the Dynamic Feedback Loop

The intervention reduces the per-pair infection probability from 2.5% to **0.5%** per period. Because the probability that a susceptible avoids infection depends exponentially on the number of infectives, **the effect propagates dynamically**. Fewer new infections can mean fewer infectives in later periods; fewer infectives increase the chance that remaining susceptible residents avoid infection; and a larger surviving population continues to contribute tax revenue.

The resulting benefit is therefore not adequately described as a simple 80% reduction in the conditional infection parameter. **The simulation captures how that parameter change interacts with the evolving state of the system.**

6.3 Why the Infection-Rate Result Is Not a Contradiction

At first glance, a 99.352% cumulative infection rate under protection may appear inconsistent with a much lower population die-out probability. It is not. The infection-rate measure asks whether residents have ever been infected by period 100; population die-out asks whether all 100 residents have died. In the model, **protection can delay transmission and mortality enough to keep residents alive longer even if most are eventually infected.** The two measures capture different dimensions of system performance.

7. Managerial and Business Implications

First, intervention evaluation should use a portfolio of outcomes rather than a single endpoint. If a decision maker considered only disease extinction, the scenarios would look nearly identical. Adding population survival and cumulative economic output changes the managerial interpretation substantially.

Second, economic value in dynamic systems can arise from timing and preservation. The protection scenario generates approximately $4.531 million more cumulative tax revenue per simulated village, not because the tax rate changes, but because more residents remain alive and economically present for more periods. This illustrates how operational or health interventions can create indirect economic value through state preservation.

Third, uncertainty should be reported explicitly. Point estimates alone can overstate precision. The confidence intervals and convergence diagnostics provide a more disciplined basis for comparing scenarios and are consistent with recommended simulation-output reporting practices that pair summary statistics with measures of statistical precision and graphical analysis [1,2].

Fourth, a policy decision would require costs as well as benefits. The workbook quantifies the gross tax-base preservation associated with protection but does not include the cost of providing or mandating the intervention. Therefore, the $4.531 million difference is not a net present value, welfare gain, or cost-benefit result. A decision-ready extension would introduce intervention cost, discounting if periods represent years, and potentially broader economic and health outcomes.

8. Limitations and Extensions

The model is intentionally stylized. It assumes homogeneous residents, a constant encounter probability, fixed infection and mortality probabilities, and a closed population. It does not include age, heterogeneous susceptibility, recovery, immunity, births, migration, capacity constraints, behavioral adaptation, treatment, vaccination, network structure, or time-varying contact rates. These simplifications make the stochastic logic transparent but limit real-world epidemiological interpretation.

The protection scenario assumes universal use and a fixed reduction in conditional infection probability. It does not model compliance, intervention failure beyond the specified probability, adoption dynamics, or intervention cost. The fiscal analysis counts gross cumulative tax from living residents and does not model income heterogeneity, unemployment, healthcare spending, productivity losses, or discounting.

The Monte Carlo analysis quantifies stochastic uncertainty conditional on the chosen input parameters. It does not quantify input uncertainty—that is, uncertainty about whether 0.05, 0.50, 0.10, and 0.10 are the correct real-world probabilities. Simulation research distinguishes output uncertainty from uncertainty in the input model; a more advanced extension could perform sensitivity analysis or assign distributions to uncertain parameters.

Several extensions would make the project richer while preserving its current logic: sensitivity analysis over transmission and mortality parameters; partial protection adoption; an explicit recovered state; alternative policy costs; expected net fiscal benefit; time-to-extinction distributions; and scenario analysis across different initial infection levels. These extensions would shift the project from a controlled teaching model toward a broader policy-analysis framework.

9. Conclusion

This project uses Excel to implement a transparent **stochastic dynamic model of infectious-disease transmission** and to **evaluate a protection intervention through repeated Monte Carlo simulation**. The model follows **susceptible**, **infected**, and **cumulative-death states** for 100 periods and summarizes 1,000 simulated villages with probability estimates, confidence intervals, economic outcomes, and convergence diagnostics.

The final workbook results show that **protection** reduces the estimated probability of **complete population die-out** from 99.3% to **56.0%**, while **disease extinction** remains near 100% in both scenarios. **Cumulative infection** is also near universal, indicating that **the intervention's principal modeled value is the preservation of population over time** rather than complete prevention of infection. That preservation translates into an estimated **$4.531 million increase** in **average cumulative tax revenue** per village, a 35.9% increase relative to no protection.

More broadly, the project demonstrates how **simulation** can reveal consequences that are hidden by terminal averages or deterministic reasoning. By combining dynamic state transitions, random variables, repeated trials, confidence intervals, convergence analysis, and economic interpretation, the workbook provides a compact example of simulation as a decision-analytics tool.

References

[1] Wieland, J. R., & Nelson, B. L. (2009). A four-level approach to reporting simulation output. Proceedings of the Winter Simulation Conference.

[2] Law, A. M. (2023). Statistical analysis of simulation output data: The practical state of the art. Proceedings of the Winter Simulation Conference.

[3] Robinson, S. (2015). A tutorial on conceptual modeling for simulation. Proceedings of the Winter Simulation Conference.

Appendix A. Key Workbook Outputs Used in This Report

All numerical findings in the report are taken from the final portfolio workbook. Because the model uses RAND(), recalculation can generate a new Monte Carlo sample. The figures and tables in this report correspond to the values stored/displayed in the workbook version used to prepare the report.

Metric	No Protection	Protection	Workbook unit
Population die-out	0.993	0.56	Probability
Disease die-out	0.993	0.996	Probability
Cumulative infection rate	1.0	0.99352	Share of original population
Average cumulative tax	12609.66	17141.03	$000s